

Business Requirements Document

Payce

Buy Now, Pay Later (BNPL) Data Platform

End-to-End Data Engineering Pipeline | Databricks + AWS

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1. Executive Summary

Payce is a Buy Now, Pay Later (BNPL) financial technology platform that allows customers to split purchases into manageable instalments. As the business scales, Payce requires a reliable, governed, and analytics-ready data platform to support decision-making across fraud prevention, credit risk, and customer management.

This document defines the business requirements for the Payce data platform - an end-to-end pipeline built on Databricks and AWS following the Bronze–Silver–Gold Medallion Architecture. The platform ingests raw financial data from multiple sources, processes and curates it into trusted datasets, and delivers business-ready data marts that answer the key questions Payce must address to operate safely and profitably.

The objective is to convert raw, inconsistent source data into a structured analytical foundation that enables the business to detect fraud, assess credit risk, and understand its customer base.

2. Business Background and Context

BNPL providers operate in a high-volume, high-risk environment. Every customer interaction generates data across the full lifecycle:

1. Application - a customer requests to split a purchase into instalments, and Payce must assess affordability.
2. Payment - the customer transacts at checkout, and funds move in real time.
3. Fraud screening - Payce must detect and block fraudulent activity.
4. Repayment - customers repay their instalments over time, or default.

Currently this data is fragmented across disparate sources with inconsistent formats, data quality issues, and no unified analytical model. Without a consolidated platform, the business cannot reliably measure fraud exposure, quantify credit risk, or segment its customers - all of which are essential to a sustainable BNPL operation.

3. Problem Statement

Payce lacks a centralised, trustworthy data platform to transform raw operational data into actionable business insight. Specifically:

- Source data arrives in multiple formats with quality issues (mixed data types, malformed records, free-text contamination, and multi-file inconsistencies).
- There is no single, governed source of truth for transactions, loans, or customers.
- Business teams cannot easily answer core questions about fraud rates, default risk, or customer segmentation.
- There is no analytical data model optimised for business intelligence and reporting.

4. Project Objectives

#	Objective
1	Ingest raw financial data from multiple sources into a centralised cloud data lake.
2	Cleanse, validate, and standardise the data into trusted, conformed datasets.
3	Model and build business-ready data marts using dimensional (star schema) design.
4	Enable analytics across three priority domains: fraud, credit risk, and customer profile.
5	Apply data governance, security, and access control throughout the platform.
6	Produce a documented, version-controlled, and reproducible data pipeline.

5. Project Scope

5.1 In Scope

- Ingestion of four source datasets representing the BNPL business lifecycle.
- A three-layer Medallion Architecture (Bronze, Silver, Gold).
- Data quality handling, type casting, and curation in the Silver layer.
- Three Gold-layer data marts (Fraud, Credit Risk, Customer) using star schema modelling.
- Data governance via Unity Catalog and secure access to cloud storage.
- Documentation: data model diagrams, this BRD, and a technical README.

5.2 Out of Scope

- Real-time streaming dashboards and production BI tool deployment.
- Machine learning model development and deployment.
- Integration with live operational/transactional systems.
- Customer-facing application development.

6. Stakeholders

Stakeholder	Interest in the Platform
Fraud Analysts	Identify fraudulent transactions and high-risk patterns.
Credit Risk Team	Assess borrower default risk and affordability.
Customer Insights Team	Understand customer segments and behaviour.
Data Engineering	Build and maintain the pipeline and data quality.
Compliance & Governance	Ensure secure, governed, and auditable data handling.
Executive Leadership	Use insights to guide strategy and manage business risk.

7. Data Sources

The platform integrates four real-world financial datasets, each mapped to a stage of the BNPL business lifecycle.

Source Dataset	Represents (BNPL Context)	Approx. Volume
PaySim	Checkout payments and transaction fraud signals	6.3 million rows
IEEE-CIS Fraud Detection	Fraud detection features (card, device, email)	590 thousand rows
Home Credit Default Risk	Customer credit applications and affordability	307 thousand rows
LendingClub	Loan repayment performance and default outcomes	2.26 million rows

8. Functional Requirements

ID	Requirement
FR-01	Ingest raw data files from cloud storage into a Bronze layer without modification, preserving full history.
FR-02	Cleanse and standardise data in the Silver layer, including correct typing, deduplication, and removal of malformed records.
FR-03	Curate wide source datasets down to business-relevant columns.
FR-04	Derive reusable business attributes (e.g. customer age, balance change, loan outcome).
FR-05	Build a Fraud data mart enabling analysis of fraud rates by transaction type and time.
FR-06	Build a Credit Risk data mart enabling analysis of default rates by grade, purpose, and term.
FR-07	Build a Customer data mart enabling segmentation by risk, income band, and education.
FR-08	Each data mart shall follow a star schema with a central fact table and supporting dimensions.
FR-09	Register all curated tables in a governed catalog for structured access.

9. Non-Functional Requirements

ID	Requirement
NFR-01	Security - Cloud credentials shall be stored securely and never exposed in code.
NFR-02	Governance - All data shall be governed through a unified catalog with access control.
NFR-03	Reliability - Ingestion shall be resilient to malformed data and reprocessing.
NFR-04	Maintainability - Code shall be version-controlled and organised by pipeline stage.

ID	Requirement
NFR-05	Scalability - Handle datasets ranging from thousands to tens of millions of rows.
NFR-06	Reproducibility - The pipeline shall be documented so it can be understood and rebuilt.

10. Solution Overview - Medallion Architecture

The platform is built on the three-layer Medallion Architecture:

Bronze Layer (Raw): Data is ingested exactly as received from source and stored as durable Delta tables. This preserves a complete, unmodified history.

Silver Layer (Cleansed): Data is cleaned, typed, validated, deduplicated, and curated to business-relevant columns. Reusable derived attributes are added. This layer is the trusted, conformed source of truth.

Gold Layer (Business-Ready): Curated Silver data is modelled into dimensional star schemas, producing the three data marts that directly serve business questions and reporting.

11. Gold Layer - Data Marts

11.1 Fraud Mart

Business question: Where, when, and how is fraud occurring across transactions?

- Fact: fct_transactions (transaction-level measures and fraud flags)
- Dimensions: dim_transaction_type, dim_date

11.2 Credit Risk Mart

Business question: Who is likely to default, and what drives credit risk?

- Fact: fct_loans (loan-level measures and outcomes)
- Dimensions: dim_grade, dim_purpose, dim_term

11.3 Customer Mart

Business question: Who are our customers, and how do they segment by risk?

- Fact: fct_customer_profile (one row per customer)
- Dimensions: dim_segment, dim_income_band, dim_education

12. Key Business Questions Answered

Domain	Example Questions
Fraud	Which transaction types have the highest fraud rate? When does fraud spike?
Credit Risk	What is the default rate by risk grade, loan purpose, or term length?
Customer	How do customers distribute across risk segments and income bands?

13. Assumptions

- Source datasets are representative proxies for Payce's BNPL business activity.
- Data volumes are sufficient to demonstrate realistic analytical scenarios.
- The platform is built for analytical and demonstration purposes, not live production operations.

14. Constraints

- The platform operates within the capabilities of the Databricks Serverless (Free Edition) environment.
- Source data quality is limited by the original public datasets.
- No live or real-time operational system integration is included.

15. Success Criteria

The project will be considered successful when:

5. All four source datasets are ingested, cleansed, and curated through the Medallion layers.
6. Three star-schema data marts are built and queryable.
7. Each data mart returns accurate answers to its core business questions.
8. The platform is governed, secured, documented, and version-controlled.

16. Data Assumptions and Limitations

The following characteristics of the source data are documented for transparency, as they inform how figures in the platform and dashboard should be interpreted.

Currency and monetary scale. The Home Credit dataset records income and credit amounts in an anonymised foreign currency, so the raw values are large relative to typical GBP figures (average recorded income is approximately 150,000+ units). The pipeline deliberately preserves these source values without rescaling, in line with the principle that the data layers should not silently alter source data. For presentation, monetary values are labelled in GBP for readability, and customers are segmented by relative income percentile rather than by absolute value. The risk relationships therefore remain valid; the absolute pound figures are illustrative rather than precise. The PaySim fraud amounts are similarly large, reflecting that synthetic fraudulent transactions in that dataset tend to drain full account balances.

Representative data. The four source datasets are real-world financial data used as representative proxies for Payce's BNPL activity. They are suitable for demonstrating the platform's analytical capability but are not live operational data.

Customer segmentation. Customer risk segments are derived from relative income bands rather than a predictive credit model. Analysis shows income alone is a weak predictor of default in this data (default rates are broadly similar across income tiers), which is reported as a genuine finding rather than adjusted.

17. Glossary

Term	Definition
BNPL	Buy Now, Pay Later - short-term instalment financing at point of sale.
Medallion Architecture	A layered data design pattern: Bronze (raw), Silver (cleansed), Gold (business-ready).
Fact Table	A central table holding measurable business events and numeric measures.
Dimension Table	A descriptive table providing context to slice and analyse facts.
Star Schema	A dimensional model with a central fact table linked to surrounding dimensions.
Default	Failure to repay a loan or instalment obligation.
Surrogate Key	A system-generated unique identifier used to link facts to dimensions.